Yonghao Shi

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Ph.D. Student in Computer Science, Simon Fraser University
Human-Computer Interaction | Smart Environments | Computational Materials | Applied AI

PROFILE

My research lies at the intersection of Human-Computer Interaction, smart environments, computational materials, sensing systems, and applied AI. I design hardware-enabled interactive systems that turn everyday materials, especially wood panels, furniture, and room-scale structures, into sensing and actuation interfaces. My current work focuses on sustainable computational wood, battery-free wireless sensing, RFID-augmented environments, low-barrier fabrication, and user-centered evaluation. Before joining SFU, I worked as a software engineer and project manager, which shaped my ability to build reliable systems and manage research prototypes end to end.

HONORS & AWARDS

- Computing Science Travel Award (CSTA)
- PhD Research Scholarship
- CMPT Grad. Fellowship

PUBLICATION

[C5] **EcoVibWood: Lower Impact Vibration Sensing in Plywood.**
Yonghao Shi, Suyuan Liu, and Xing-Dong Yang.
ACM Symposium on User Interface Software and Technology (UIST 2026).

[C4] **3Duino: A Low-Barrier Platform for Prototyping Interactive 3D-Printed Devices.**
Yonghao Shi, Zejia Cai, Liuyi Chen, Yuning Su, Liang He, Xing-Dong Yang, and Te-Yen Wu.
The 10th Annual ACM Symposium on Computational Fabrication (SCF 2025). DOI: 10.1145/3745778.3766649

[C3] **Woodowel: Electrode Isolation for Electromagnetic Shielding in Triboelectric Plywood Sensors.**
Yonghao Shi, Chenzheng Li, Yuning Su, Xing-Dong Yang, and Te-Yen Wu.
Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI 2024). DOI: 10.1145/3613904.3642304

[C2] **Tagnoo: Enabling Smart Room-Scale Environments with RFID-Augmented Plywood.**
Yuning Su, Tingyu Zhang, Jiuen Feng, Yonghao Shi, Xing-Dong Yang, and Te-Yen Wu.
Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI 2024). DOI: 10.1145/3613904.3642356

[C1] **Laser-Powered Vibrotactile Rendering.**
Yuning Su, Yonghao Shi, Da-Yuan Huang, and Xing-Dong Yang.
Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct 2023). DOI: 10.1145/3586182.3615795

In submission

- Lumberina: A Wireless Smart Wood Panel with Concealed Dot-Matrix Display (in press)
- Development and Evaluation of Sensor-Powered Wireless Smart Wooden Panels (in press)

WORK EXPERIENCE

Sep. - Dec. 2026 — **Research Intern, Huawei, HMI Lab, Toronto, Canada.**
Four-month internship: Sep. 1 - Dec. 31, 2026 (expected end date).

Sep. 2020 - Nov. 2021 — **Project Manager, Institute of Information Engineering, Chinese Academy of Sciences.**
Managed software and data-oriented engineering projects, coordinated timelines, and supported cross-functional delivery.

Nov. 2018 - Sep. 2020 — **Software Engineer, Institute of Information Engineering, Chinese Academy of Sciences.**
Worked on backend systems, data infrastructure, and software development for research and engineering applications.

Jun. 2017 - Sep. 2017 — **Software Engineer, PwC Beijing.**
Contributed to software engineering tasks in a professional consulting and technology environment.

EDUCATION

Sep. 2022 - Present — **Ph.D. in Computer Science, Simon Fraser University, Burnaby, Canada.**
Research: Human-Computer Interaction, smart environments, and computational materials.

Sep. 2017 - Sep. 2018 — **M.Sc. in Computer Science, University of Plymouth, United Kingdom.**
Thesis: Data Science and Business Analysis.

Sep. 2016 - Jun. 2017 — **B.Sc. in Electrical and Electronic Engineering, University of Plymouth, United Kingdom.**

Sep. 2013 - Jun. 2016 — **B.Sc. in Electrical and Electronic Engineering, Beihang University, China.**